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Potter

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(54) **HANDHELD HAIR PRODUCT DISPENSER**

(71) Applicant: **Alanna Potter**, Los Angeles, CA (US)

(72) Inventor: **Alanna Potter**, Los Angeles, CA (US)

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A45D 34/04 (2006.01)
A46B 11/00 (2006.01)
A46D 1/00 (2006.01)

(52) **U.S. Cl.**
CPC ***A45D 24/28*** (2013.01); ***A45D 34/042*** (2013.01); ***A46B 11/0006*** (2013.01); ***A46D 1/0246*** (2013.01); *A45D 2200/05* (2013.01); *A45D 2200/055* (2013.01); *A46B 2200/104* (2013.01)

(58) **Field of Classification Search**
CPC A45D 24/28; A45D 24/26; A45D 24/22; A45D 19/026; A46B 11/0031; A46B 11/0034; A46B 11/0027; A46B 11/0055; A46B 11/0051
USPC 132/114, 115
See application file for complete search history.

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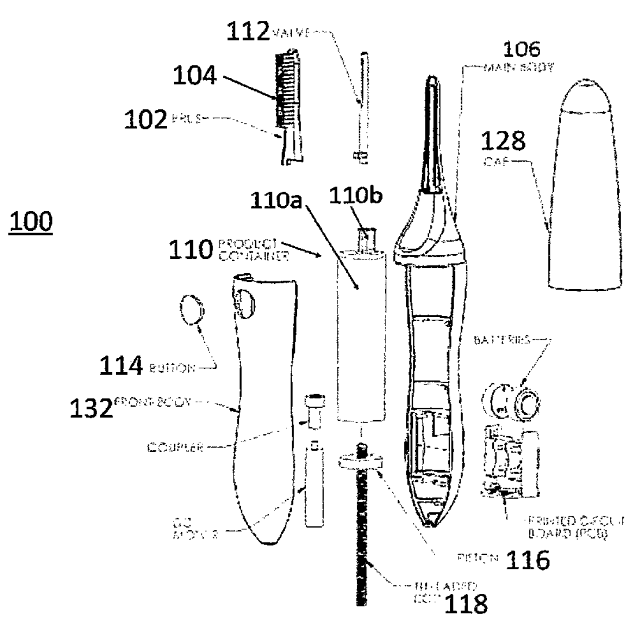
(57) **ABSTRACT**

A dispensing brush and/or comb assembly is made to contain and selectively dispense a fluid. The assembly includes a brush and/or comb having an internal cavity for accepting a fluid, hollow bristles, and a body couplable to the brush and/or comb, having an internal space for receiving a fluid container. The fluid container is removably placed within the internal space of the body and is fluidly connected to the brush and/or comb internal cavity in an installed position, and a cover for retaining the fluid container within the body.

13 Claims, 3 Drawing Sheets



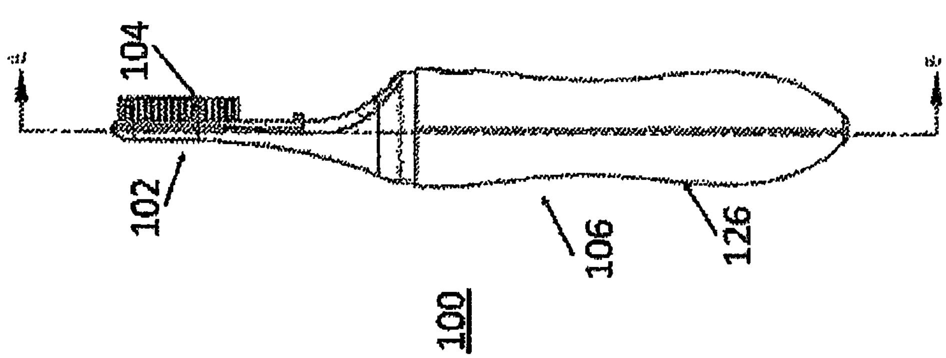


FIG. 1

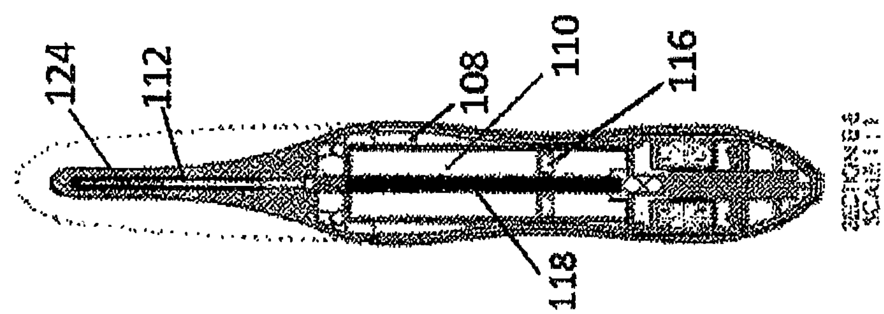


FIG. 2

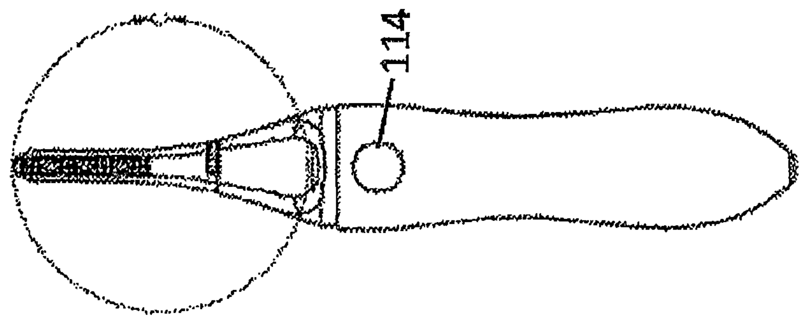


FIG. 3

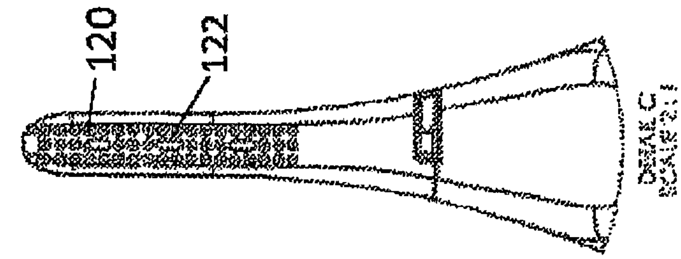
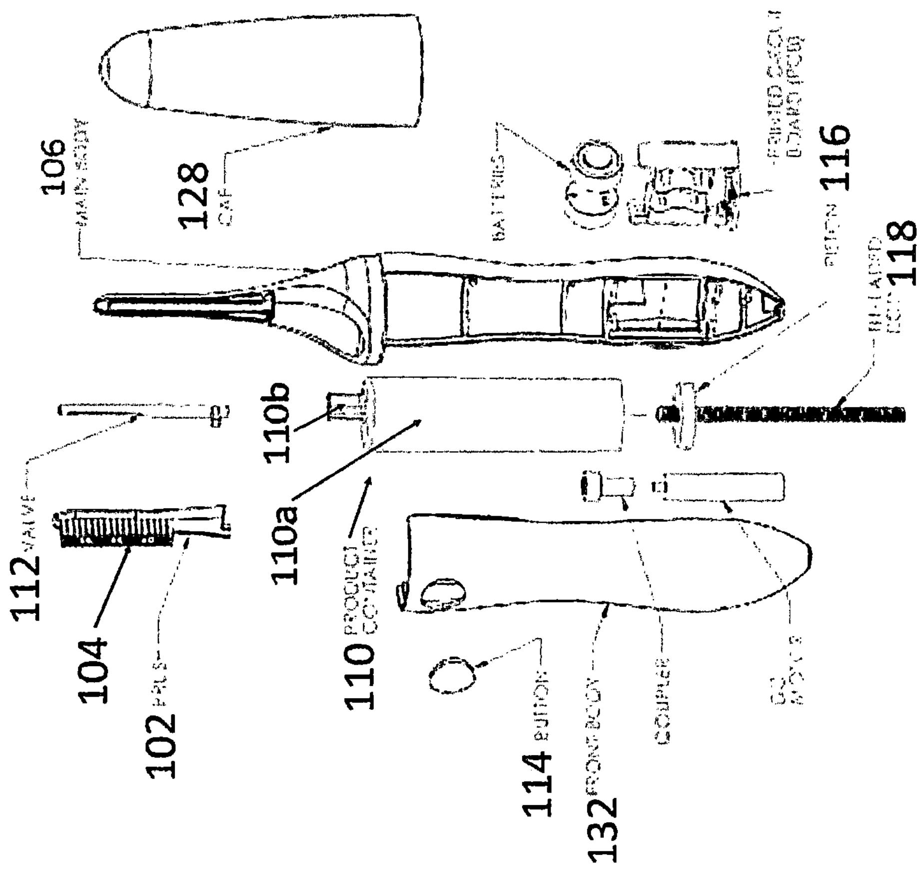


FIG. 4


106
MAIN BODY
128
CAP
BATTERIES
PRINTED CIRCUIT BOARD (PCB)
116
PISTON
118
THREADED ROD
112
VALVE
110b
110a
104
102
110
PRODUCT CONTAINER
114
BUTTON
132
FRONT BODY
COUPLER
DC MOTOR

100

FIG. 5

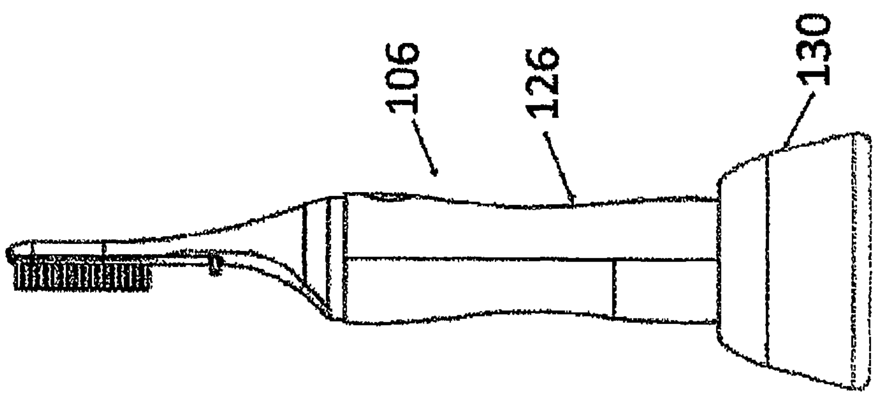


FIG. 6

HANDHELD HAIR PRODUCT DISPENSER

BACKGROUND

Technological Field

The disclosure relates to a brush for comb device having a selectively dispensing a hair care product, and more specifically the brush or comb device having a removable dispensing container allowing the user to replace or refill the dispensing unit when it is empty, or to dispense a different product altogether.

Description of Related Art

Innumerable variations of combs, hairbrushes, and the like have been developed in the past. Many such devices have been developed for some specific purpose, i.e., to solve some problem in the field of grooming and hair care. While routine, day-to-day hair care, i.e., combing and brushing, is generally a single step process requiring the use of only a single tool or device (comb, brush, etc.), the need or desire for more involved hair treatment may require several steps.

An example for treatment is the process of minimal dying for grays, streaks of coloring for the hair, and putting other products in, like hair relaxers and edge control to tame hairline flyaways. Many people wish to change the color of their hair, perhaps as a change of pace, or to go with a new wardrobe, a change of the season, or for a special occasion. While other brushes developed in the past have been used for one particular purpose, the present disclosure allows a user to change between products and product types that are used by the brush.

SUMMARY

A dispensing brush and/or comb assembly meant to contain and selectively dispense a fluid is disclosed. The assembly includes a brush and/or comb having an internal cavity for accepting a fluid,

a body couplable to the brush and/or comb, having an internal space for receiving a fluid container. The fluid container is removably placed within the internal space of the body and is fluidly connected to the brush and/or comb internal cavity in an installed position. The assembly can include a charging stand having a cavity for accepting a portion of the body.

The fluid container can be coupled to the brush and/or comb by a valve located within the body.

The fluid container can include a main section and a narrowed section for connecting to the valve.

The body can include a switch to activate a piston configured to force the fluid from the fluid container into the internal cavity of the brush and/or comb. The switch can be battery powered. The piston can be coupled to a threaded rod. The threaded rod can extend through the length of the fluid container. The piston can be actuated by a DC motor.

The brush and/or comb can include a plurality of bristles wherein each of the bristles of the plurality of bristles are parallel to each other. The brush and/or comb can be lockably coupled with the body. The body can extend along the length of the brush and/or comb.

The bristles do not extend past an outer surface of the body. A cover can be removably coupled to the body over the brush and/or comb. The brush and/or comb can include at least one dispensing nozzle. The valve can include a plurality of holes on a first side emanating perpendicular to the longitudinal axis of the valve configured to mate with the at least one dispensing nozzle.

The brush and/or comb can be configured to turn the assembly "on" in a first position and to turn the assembly "off" in a second position. The brush and/or comb can be rotatable between the first position and the second position In the second position an outlet of the valve can face away from a dispensing nozzle of the brush and/or comb. The body can include an undulating outer surface.

These and other features of the systems and methods of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description of the preferred embodiments taken in conjunction with the drawings.

FIGURES DESCRIPTION

The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

FIG. **1** is a side view of a dispensing brush and/or comb assembly according to an embodiment of the disclosure;

FIG. **2** is a sectional view of FIG. **1**;

FIG. **3** is a front view of FIG. **1**;

FIG. **4** is an enlarged view of section C of FIG. **3**;

FIG. **5** is an exploded view of FIG. **1**; and

FIG. **6** is a side view of the assembly of FIG. **1** in a charging stand.

DETAILED DESCRIPTION

Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a partial view of an exemplary embodiment of a hair product dispensing brush and/or comb assembly in accordance with the invention is shown in FIG. **1** and is designated generally by reference character **100**. Other embodiments of the assembly in accordance with the invention, or aspects thereof, are provided in FIGS. **2-6**, as will be described. The methods and systems of the invention can be used to use various dyes and hair products interchangeably in the same brush and/or comb or applicator.

FIG. **1** shows a dispensing brush and/or comb assembly **100** that contains and selectively dispenses a fluid as needed. The assembly **100** includes a brush and/or comb applicator **102**. The brush and/or comb applicator **102** includes an internal cavity and hollow bristles **104** (shown in FIG. **2**) through which the fluid travels. The assembly **100** includes a body **106** which couples to the brush and/or comb applicator **102**. The body **106** has a handle portion and a head portion extending longitudinally from a distal end of the handle portion. The head portion has a proximal portion which tapers gradually inwardly toward the longitudinal axis (shown in FIGS. **1** and **5**) and a distal portion to which the brush and/or comb applicator is detachably coupled. The handle portion has a rear half-shell and a front half-shell/front cover **132**, where the rear half-shell and front half-shell define an internal space **108** (shown in FIG. **2**) which receives and houses the fluid container **110** while the assembly **100** is in use. The fluid container **110** is removably placed within the internal space **108** of the body and is fluidly connected to the brush and/or comb internal cavity and hollow bristles **104** in an installed position.

FIG. **2** shows an inner working of the assembly **100**. The fluid container **110** is coupled to the brush and/or comb **102** by a valve **112**. The fluid container **110** includes a main section **110***a* and a narrowed section **110***b* for connecting to the valve **112**. FIG. **3** shows a switch **114** which is used to activate a piston **116** which drives through and decreases the inner volume of the fluid container **110** and forces the fluid from the fluid container **110** into the brush and/or comb applicator **102**. The switch or button **114** and piston **116** are shown being battery powered, but one will readily understand that other powering mechanisms and systems can be employed. The piston **116** is coupled to and actuated by a turning threaded rod **118**. A user depresses dispensing button **114** momentarily to dispense a set amount of liquid or gel. The user could also hold down the dispensing button **114** to continuously dispense liquid or gel. After depressing dispensing button **114** or while holding button the user can then apply brush and/or comb and dispense liquid or gel to their hair. The threaded rod **118** extends the length of the fluid container **110**. The piston **116** can be actuated by a DC motor or any other acceptable means. When the device has depleted the dispensing container **110**, the dispensing mechanism will stop and the device will notify the user via the light and/or sound.

When the device has dispensed all of the liquid or gel in the dispensing container **100** the user can replace the container **110** by removing a front half-shell/cover **132**. The old dispensing container can be removed and a new one inserted. The dispensing cover **132** is then replaced/reconnected.

FIG. **4** shows a close up of the brush and/or comb applicator **102**. The brush and/or comb applicator **102** includes a plurality of bristles **120**. The bristles **120** are aligned parallel to each other and point in the same direction. The brush and/or comb applicator **102** includes several dispensing nozzles **122** which are used for placing the fluid on the user's head. The valve **112** includes a plurality of holes **124** (shown in FIG. **2**) on a front side thereof, emanating perpendicular to the longitudinal axis of the valve **112** which mates with the dispensing nozzle **122**, and as the fluid travels up the valve **112** from the fluid container **110**, the fluid enters the brush and/or comb bristle nozzles **122**.

The brush and/or comb applicator **102** is lockably coupled with the body **106**. The body **106** extends all the way along the length of the brush and/or comb applicator **102**, in order to provide support and locking capabilities to the brush and/or comb applicator **102**. The bristles do not extend past an outer surface **126** of the body **106**, which allows a cover **128** to be placed over the brush and/or comb when not in use to prevent leaks from oozing out. The brush and/or comb applicator **102** is also used to turn the assembly "on" in a first position and to turn the assembly "off" in a second position. The brush and/or comb applicator **102** is rotatable between the first position and the second position. In the first position, the dispensing nozzle **122** and outlet(s) of the valve **112** are arranged to dispense the fluid from a front side of the assembly (fluid is not dispensed from a rear surface of the distal portion of the head). In the second position an outlet of the valve **112** faces away from the dispensing nozzle **122** of the brush and/or comb applicator.

As can be seen in FIG. **6**, the handle portion of the body **106** has an undulating outer surface **126** for better grip. The assembly **100** can include a charging stand **130** which has a cavity for accepting a portion of the body in order to charge is using a USB, or other type of plug. When the device runs low on battery, a light and/or sound will alert the user to the low battery status. The user can then charge the device by placing it into the base station **130**. The light and/or sound will later notify the user that charging is complete. Assembly **100** may alternately contain non-replaceable batteries surrounded in copper coils, and additional copper coils in the charging stand **130**, coupled to a PCB controller, to produce a faraday charge affect, between the interface of the two placed copper coils, and the concealed product device battery.

The methods and systems of the present disclosure, as described above and shown in the drawings provide for a hair product dispensing brush and/or comb with superior properties including increased reliability and ease of use. While the apparatus and methods of the subject disclosure have been showing and described with reference to embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the spirit and score of the subject disclosure.

The invention claimed is:

1. A hair product dispensing brush and/or comb assembly being configured to contain and selectively dispense a fluid onto hair, said assembly comprising:

a brush and/or comb applicator having a proximal end opposite a distal end, an internal cavity for accepting a fluid, a plurality of bristles extending from a front side of the brush and/or comb applicator, and at least one dispensing nozzle surrounded by the plurality of bristles,

a body having a handle portion with a proximal end and a distal end, and a head portion extending longitudinally from the distal end of the handle portion, where a longitudinal axis of the assembly extends through the proximal and distal ends of the brush and/or comb applicator and the proximal and distal ends of the body,

the head portion having a proximal portion which tapers gradually inwardly toward the longitudinal axis, and a distal portion to which the brush and/or comb applicator is detachably coupled,

the handle portion having a rear half-shell and a front half-shell where the rear half-shell and the front half-shell define an internal space

a fluid container removably placed within the internal space of the handle portion of the body and is fluidly connected to the internal cavity of the brush and/or comb applicator in an installed position, where the fluid container includes a main section and a narrowed section, and

a valve coupled to the narrowed section of the fluid container and extending longitudinally within the head portion of the body, the valve including a plurality of outlets on a front side thereof emanating perpendicular to the longitudinal axis of the valve and configured to mate with the at least one dispensing nozzle,

wherein the bristles of the plurality of bristles and an entirety of the head portion do not extend past an outer surface of the handle portion of the body,

wherein the brush and/or comb applicator is rotatable between a first position and a second position, wherein the first position the plurality of outlets of the valve face the at least one dispensing nozzle for permitting dispensing and in the second position the plurality of outlets of the valve face away from the at least one dispensing nozzle and wherein fluid is not dispensed from a rear surface of the distal portion of the head.

2. The hair product dispensing brush and/or comb assembly of claim 1, wherein the fluid container is coupled to the brush and/or comb by the valve.

3. The hair product dispensing brush and/or comb assembly of claim 1, further comprising a piston within the fluid container, wherein the body includes a switch to activate the piston to force fluid from the fluid container into the internal cavity of the brush and/or comb applicator.

4. The hair product dispensing brush and/or comb assembly of claim 3, wherein the switch is battery powered.

5. The hair product dispensing brush and/or comb assembly of claim 3, wherein the piston is coupled to a threaded rod.

6. The hair product dispensing brush and/or comb of the assembly of claim 5, wherein the threaded rod extends through the length of the main section of the fluid container.

7. The hair product dispensing brush and/or comb of the assembly of claim 5, wherein the piston is actuated by a DC motor.

8. The hair product dispensing brush and/or comb assembly of claim 1, wherein each of the bristles of the plurality of bristles are parallel to each other.

9. The hair product dispensing brush and/or comb assembly of claim 1, wherein the body extends along the length of the brush and/or comb applicator.

10. The hair product dispensing brush and/or comb assembly of claim 1, further comprising a charging stand having a cavity for accepting a portion of the body.

11. The hair product dispensing brush and/or comb assembly of claim 1, wherein a cover is removably coupled to the body over the brush and/or comb applicator.

12. The hair product dispensing brush and/or comb assembly of claim 1, wherein the body includes an undulating outer surface.

13. The hair product dispensing brush and/or comb assembly of claim 1, wherein the front body of the handle portion is detachably coupled to the rear body of the handle portion.

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